NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR

Even Semester Mid-Term Examination, 2025-26

Course Code: CEC402 — Full Marks: 25

Course Name: Design of Concrete Structures — Time: 90 minutes

Question Paper No.: NITDGP/CEC402/

Instructions: Answer per instructions. Materials to be supplied: IS: 456 – 2000.

PART A

Question No.	Body of the Question **(ANSWER ANY ONE USING M25 AND FE500)**	Marks	Mapped CO
1	Assume a **straight line instead of a parabola** for the stress-strain curve for concrete with a partial safety factor of 1.6 for concrete and 1.2 for steel. A rectangular under-reinforced concrete section of 250 mm width and 400 mm effective depth is reinforced with 4 nos. 25 mm tor Φ bars. **Find from the above condition and deviation from actual:** a). The depth of N.A., b). Moment of resistance c). What will be the UDL that the beam can carry in actual conditions, without self-weight? d). Check the adequacy of the rebar diameter if the effective span of the beam is 5.25 m, width of the support is 250 mm	(3 +1) + (2 +1) + 2.5 + 3 = 12.5	CO1 & CO2
OR	Design the beam with the following parameters: Clean span 5.50 m, width of support 250 mm, external load 22 kN/m	12.5	

PART B

Question No.	**Body of the Question**	Marks	Mapped CO
1	Design a circular column with helical reinforcement subjected to an axial load of 1000kN under service load and live load. The column has an unsupported length of 3.6m and is effectively held in position at both ends but restrained against rotation at one end. Use M25grade of concrete and Fe415 steel.	12.5	CO3

CO1: Apply knowledge of solid mechanics for design solutions.

CO2: Understand basic design philosophies applicable to concrete structures.

CO3: Formulate, analyse, and design basic components of Civil Engineering Reinforced Concrete structures.

$\boxed{0.87 f_y \cdot A_{st} \cdot z_u}$

M_{ur}

$0.36 f_{ck} \cdot x_{ub} \cdot z_u$

$\frac{\pi}{4}\phi^2 = 1000$

$\phi = \sqrt{\frac{4000}{\pi}}$

$\frac{M_{ulim}}{0.36 f_{ck} x_{ub} \cdot z_u}$

$M_{ulim} = 0.133 f_{ck} b d^2$

$A_{st} = \frac{M_u}{0.87 f_y z_u}$

$D = \frac{l_{eff}}{\ldots}$

322.72

NITDGP/B.Tech/ Regular/Even/2025-26

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR

Even Semester End-Term Examination, 2025-26

Course Code: CEC402 | **Full Marks: 60**

Course Name: Design of Concrete Structures | **Time: 180 minutes**

Question Paper No.: NITDGP/CEC402/

Instructions: Answer any **FOUR**. Before starting the exam, IS: 456 – 2000 is required.

PART A

Q. No.	Body of the Question **(Adopt M25 & FE415)**	Marks	Mapped CO
1	A beam is 6 x 0.25 x 450 m^3, subjected to a 15 kN/m service live load, moderate exposure with 2 hrs fire resistance, and 3 nos 20 mm tor phi tension reinforcement. a). Determine the depth of the neutral axis. $\frac{Bx^2}{2} = mA_{st}(d - x/a)$ $m = \frac{280}{3\sigma_{cbc}}$ b). Determine the liver arm. $\rightarrow (d - \frac{x}{3})$ c). Determine the moment of resistance. If 2 nos 16 mm tor phi rebars are used as compression reinforcement, then d). Calculate the moment of resistance.	3 + 3 + 3 + 6 = 15	CO1 + CO2
2	Design a section of a ring beam 500 mm wide and 700 mm deep, subjected to a bending moment of 200 kNm, twisting moment of 15 kNm and s shear force of 150 kN at ultimate.	5 x 3 = 15	CO3
3	A 12 x 0.60 x 0.4 m^3 intermediate beam, 1 year old, carries an all-inclusive service load of 22 kN/m, of which 60% is permanent, severe exposure with 2 hrs fire resistance. a). Calculate the final deflection at mid span of the beam after one year from the date of construction. b). Calculate the crack width at mid span of the beam at the bottom corner	4 + 8 + 3 = 15	CO1 + CO2
4	Design a column subjected to an axial load of *2000 kN* under service dead load and live load. The column has an unsupported length of *4.0 m* and effectively held in position and restrained against rotation at both ends.	15	CO3
5	Design an isolated footing for a square column, *400 mm × 400 mm* with *12 – 20 mm* diameter longitudinal bars carrying service loads of *1200 kN*. The safe bearing capacity of soil is *250 kN/m^2* at a depth of *1 m* below the ground level.		CO3

CO1: Apply knowledge of solid mechanics for design solutions.

CO2: Understand basic design philosophies applicable to concrete structures.

CO3: Formulate, analyse, and design basic components of Civil Engineering Reinforced Concrete structures.